

Abelardo Diaz Sanchez

Full-Stack Developer & AI Agent Architect

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PROFILE

Full-stack developer specialized in AI agents built on Claude. I design, build, and operate production systems: multichannel conversational agents (WhatsApp, Telegram, web) with RAG and guardrails, multi-tenant SaaS, synced ecommerce, and integrations with real hardware (CCTV, GPS). Everything runs on infrastructure I manage myself: Linux VPS, PostgreSQL, Docker, PM2, Cloudflare. Before code: 10 years founding and operating REDV6, a WISP serving rural communities in San Luis Potosi, Mexico, 24/7. Verifiable portfolio with case studies: abelardodiaz.dev.

SELECTED PROJECTS

Agente 080

2026

Multichannel AI agent (WhatsApp, Telegram, web) that goes beyond replying: it reasons, decides, and executes real business actions — collections, sales, support, scheduling, real-estate inventory — with multiple specialist agents, memory, RAG, and guardrails. Bounded autonomy: every action has defined limits and is auditable.

[Python](#) [FastAPI](#) [Celery](#) [Redis](#) [PostgreSQL](#) [Claude API](#)

Internal Multi-AI Platform + Project Manager

internal use, in production

2025-2026

Two APIs that govern my operation: (1) a unified gateway over 8 AI providers (OpenAI, Gemini, DeepSeek, Grok, Groq, Qwen, Kimi + web search) with persistent conversations, multi-AI debates with automatic consensus, a knowledge base that intercepts queries, per-project auth with key rotation, and an MCP server consumable from Claude Code; (2) a centralized project manager: fleet-wide catalog and health checks, git progress tracking, Cloudflare management (DNS/SSL/cache) with granular per-key permissions, and RAG on pgvector (hybrid vector + full-text search with RRF fusion).

[Python](#) [PostgreSQL + pgvector](#) [MCP](#) [REST APIs](#) [Cloudflare API](#)

citas-bot-universal

open source, MIT

2026

Python template for WhatsApp appointment assistants built on Claude: FastAPI + SQLite + APScheduler. Distilled from a private production system for medical practices; ready to fork and deploy.

[Python](#) [FastAPI](#) [Anthropic API](#) [WhatsApp Business API](#)

Video Frame Analyzer

2026

CLI that extracts video frames with FFmpeg scene detection, analyzes them with Gemini Flash (vision), and generates a narrative with Claude. Multi-model pipeline at ~\$0.008 USD per video.

[Python](#) [FFmpeg](#) [Gemini Vision](#) [Claude](#) [Pydantic](#)

Domus Rentas: multi-tenant SaaS

2025-2026

Evolution of a production rental platform into a multi-tenant SaaS built as 9 polyglot microservices: Django where it fits, FastAPI where real time matters, and a Next.js frontend acting as gateway.

[Django](#) [FastAPI](#) [Next.js](#) [PostgreSQL](#) [Redis](#) [Docker](#)

Tienda Django Sync

2026

Ecommerce whose real engine is not the storefront: automatic sync of catalog, prices, inventory, and exchange rate against the Syscom wholesaler API.

Django Next.js Celery PostgreSQL Docker

ulfbk - Recyclable Blocks open source

2026

Ecosystem of 28 composable multi-stack packages (Python + TypeScript): reusable infrastructure for building APIs, multitenant SaaS, bots, and dashboards. Monorepo with uv + pnpm + Turborepo, zero circular dependencies.

Python TypeScript FastAPI Next.js Turborepo

Call Blocker open source, published on F-Droid

2026

Android app for blocking unwanted calls, published on F-Droid (passes the store's reproducibility and licensing review). Kotlin + Jetpack Compose + Clean Architecture.

Kotlin Jetpack Compose Material 3 Hilt Room

EXPERIENCE

Full-Stack Developer & AI Agent Architect — Independent

2024 – present

I build and operate production systems for clients in Mexico and Europe with an AI-native workflow: spec-driven development with Claude Code as my primary tool, reusable skills and commands portable across projects, persistent memory, MCP integrations (own servers and clients), and orchestration of 8 AI models by cost and context. Full cycle: architecture, implementation, deployment, and operations on my own servers, with a secure publication pipeline (gitleaks + custom sanitization). Beyond the selected projects: video surveillance backends (Hikvision ISAPI, 19 REST endpoints) and GPS fleet tracking (Tracksolid, 26 endpoints), verified against real hardware from my own operation.

Founder & Technical Director — REDV6 (WISP)

2015 – present

Wireless ISP bringing connectivity to rural communities in San Luis Potosi underserved by major carriers. 24/7 operations with residential and SMB customers.

- Design and operation of long-distance point-to-point links at 2.4/5/60 GHz (Cambium, MikroTik, Altai).
- Enterprise networks, high-coverage WiFi, IP PBX systems, CCTV, and GPS fleet tracking.
- Off-grid solar backup power systems for remote sites.
- Today it serves as the real-hardware lab where I verify my software integrations.

Sales Executive — AT&T Mexico

2013 – 2015

Consultative sales of technology and internet services to businesses.

Sales Executive — Axtel Telecomunicaciones

2010 – 2013

Consultative sales of technology and internet services to businesses.

Digital Marketing & Online Business Consultant — Consultoria e Internet para Empresas (WSI franchise)

2005 – 2010

Digital marketing consulting and online business development for SMBs under the international WSI franchise methodology.

IT & Networking Instructor — Tec. en Sistemas (Salinas de Hidalgo) / Inst. Quetzal (Zacatecas)

2001 – 2003

Taught LAN networking, Windows 2000 Server, and Linux (Red Hat 8, Mandrake 9); PC assembly and maintenance.

SKILLS

Languages	Python, TypeScript, SQL, Kotlin (Android)
AI & agents	Claude API (agents, tool use), Claude Code, MCP (servers and clients), RAG (pgvector, hybrid search), multi-provider orchestration (8 models), spec-driven development, Whisper, Gemini Vision, context engineering
Backend	FastAPI, Django, Flask, Celery, Redis, PostgreSQL, WhatsApp Business API, webhooks (HMAC, retry queues), Stripe
Frontend	Next.js, React, Tailwind CSS, Astro
Infra / DevSecOps	Linux (self-hosted VPS), Docker, PM2, Cloudflare, Vercel, GitHub Actions, gitleaks + pre-publication sanitization, MikroTik/Cambium networking

CERTIFICATIONS

MTCNA - Network Associate	MikroTik, 2018	MTCRE - Routing Engineer	MikroTik, 2018
ePMP - Wireless links	Cambium Networks, 2018	Super-coverage WiFi	Altai Technologies, 2018

EDUCATION & LANGUAGES

Computer Systems Engineering	— Instituto Tecnológico de Zacatecas (partial coursework)	2000 – 2004
Spanish: native · English: technical — fluent reading, working conversation.		